

in the rapidly changing fetal environment. Additionally, both in vitro and in vivo responses to cytokine growth factors may help to delineate which tissue sources will provide the highest rates of long-term engraftment.

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## Differential regulation of seizure activity in the hippocampus of male and female rats

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**OBJECTIVE:** The aim of the current study was to directly examine and compare the susceptibility to N-methyl-D-aspartate-induced seizures in male versus female rats. We also sought to compare the anticonvulsant effects of magnesium sulfate in these two groups.

**STUDY DESIGN:** Eighteen female and 10 male rats were stereotactically implanted with a chronic bipolar recording electrode in the hippocampus and an injection cannula in the lateral cerebral ventricle. After 1 week rats randomly received an intravenous injection of 90 mg/kg magnesium sulfate or saline solution control. Fifteen minutes after the infusion rats were given the convulsant N-methyl-D-aspartate at a dose of 5  $\mu\text{g}/\mu\text{l}$  by direct intraventricular injection. Electrical seizure activity was thereafter assessed for 20 minutes. All data were analyzed by the Mann-Whitney *U* test and Student *t* test.

**RESULTS:** In saline solution-treated rats receiving the convulsant N-methyl-D-aspartate, females had significantly lower total duration ( $p < 0.01$ ) and total number of seizures ( $p < 0.05$ ) compared with the male rats. The initial seizure was not affected by gender. In the female animals magnesium sulfate significantly reduced first seizure duration ( $p < 0.01$ ) compared with saline solution controls. In males magnesium sulfate reduced both total duration ( $p < 0.05$ ) and total seizure number ( $p < 0.05$ ) compared with saline solution-treated animals.

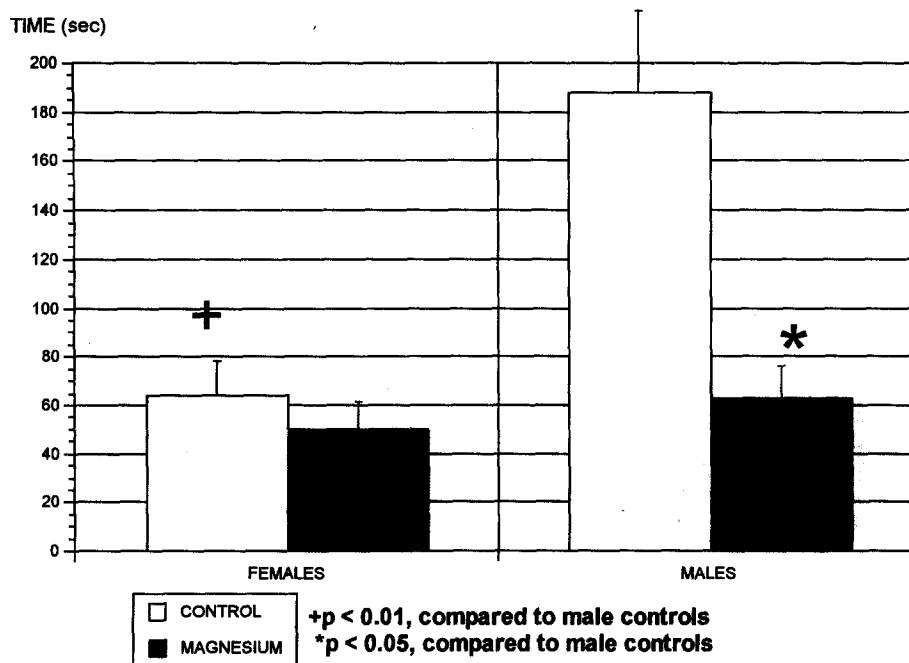
**CONCLUSION:** N-methyl-D-aspartate-induced seizure activity is more severe in males versus female rats. Magnesium sulfate's effect on N-methyl-D-aspartate-induced seizures is also dependent on gender. We speculate that seizure regulation may be hormonally influenced. (*AM J OBSTET GYNECOL* 1995;173:1160-5.)

**Key words:** Rats, magnesium, sex characteristics, seizures

Studies of seizure activity to date have focused on male animals, and treatments have been based on these results. Recent studies suggest that males and females react differently under various seizure-initiating proto-

cols. For example, female rats have demonstrated higher thresholds for pentylenetetrazol-induced seizures than have males.<sup>1</sup> On the other hand, females have lower seizure thresholds than males have for electroshock-induced seizures.<sup>1</sup> Moreover, sensitivity to the convulsant picrotoxin is dose dependent: females seize more readily at lower doses of picrotoxin, whereas males seize more at higher doses.<sup>2</sup> These studies suggest that sex hormones may influence seizure susceptibility. Sex hormones are found in brain areas that regulate sexual behavior, such as the amygdala and

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**Fig. 1.** Total seizure duration over 20-minute recording period in female and male rats given 5  $\mu$ g NMDA intraventricularly and either saline solution control or magnesium sulfate intravenously. Female controls had significantly reduced total seizure duration compared with male controls (*plus sign*,  $p < 0.01$ ). Magnesium sulfate treatment significantly reduced total seizure duration in males but not in females (*asterisk*,  $p < 0.05$  compared with male controls). Error bars, SEM.

hippocampus.<sup>3</sup> These areas are also important for the generation and maintenance of seizure activity.

Estrogen has recently been shown to enhance excitatory amino acid-induced responses.<sup>4</sup> N-methyl-D-aspartate (NMDA) is an excitatory amino acid that has been used to chemically induce seizure activity in animals. The NMDA receptor has been implicated in the production and maintenance of seizure activity<sup>5</sup> and is found in high density in the hippocampus and cortex.<sup>6</sup>

The primary objective of the current study was to compare seizure activity induced by NMDA in male versus female rats. Secondly, we sought to compare the anticonvulsant effects of magnesium sulfate, a noncompetitive antagonist of the NMDA receptor, in these two groups.

#### Material and methods

Eighteen female and 10 male Long-Evans rats (Charles River, Portage, Mich.) were individually housed in an environmentally controlled vivarium under 12-hour light/12-hour dark cycles. Food and water were available ad libitum. Female (250 to 300 gm) and male (300 to 340 gm) rats were matched for age. The Wayne State University guide to the use and care of animals was followed.

Rats were surgically anesthetized with 45 to 60 mg/kg of pentobarbital sodium intraperitoneally. A bipolar recording electrode (MS 303/2, Plastics One, Roanoke,

Va.) was stereotaxically implanted into the hippocampus according to the following coordinates: 3.5 mm posterior to Bregma, 2.0 mm lateral to midline, and 3.5 mm ventral to the surface of the skull.<sup>7</sup> In addition, a 23-gauge stainless steel cannula was stereotaxically implanted into the lateral cerebral ventricle according to the following coordinates: 2.0 mm anterior to Bregma, 1.0 mm lateral to midline, and 5.4 mm ventral to the surface of the skull.<sup>7</sup> The cannula was temporarily blocked with an obturator until the time of the experiment. The electrode and cannula assembly was anchored to the skull with three stainless steel jeweler's screws and cranioplastic cement. One of the screws served as a ground. Animals were allowed 1 week to recover from the surgical procedure.

On the day of experimentation rats were placed into a small animal restrainer and remained in the restrainer for the duration of the experiment. The restrainer allowed for isolation of electrical seizure activity. An electroencephalogram was recorded on a Grass model 7 polygraph (Grass Instrument, Quincy, Mass.) by the implanted electrode. After baseline electroencephalographic activity was recorded, rats randomly received an intravenous injection of 90 mg/kg magnesium sulfate or saline solution. This dose was chosen on the basis of a previous report showing therapeutic blood levels (i.e.,  $5.18 \pm 0.21$  mg/dl) with intravenous administration.<sup>8</sup> Fifteen minutes after the infu-

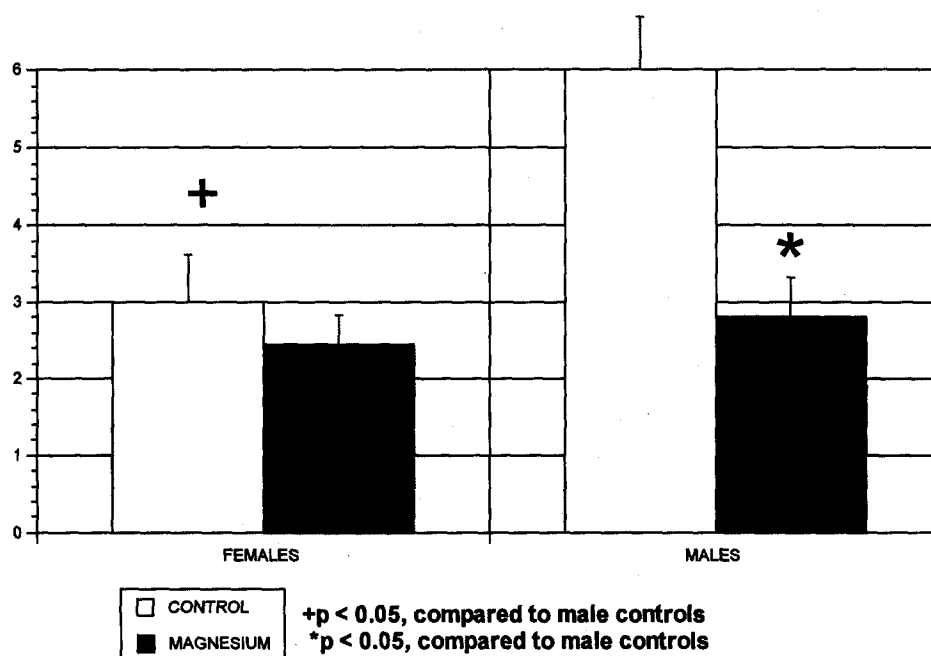


Fig. 2. Total seizure number over 20-minute recording period in female and male rats given 5  $\mu$ g NMDA intraventricularly and either saline solution control or magnesium sulfate intravenously. Female controls had significantly reduced total number of seizures compared with male controls (*plus sign*,  $p < 0.05$ ). Magnesium sulfate treatment significantly reduced total seizure number in males but not in females (*asterisk*,  $p < 0.05$ , compared with male controls). Error bars, SEM.

sion rats were given 1  $\mu$ l of 5 mg/ml NMDA through their indwelling cannulae at a rate of 1  $\mu$ l/min. The injector was left in place for 1 minute after the end of NMDA infusion and then replaced with the obturator. Seizure activity was assessed for 20 minutes from the end of the NMDA injection. Seizures were identified as repetitive epileptiform spiking lasting at least 4 seconds. Onset to first seizure, first seizure duration, total seizure duration, and total seizure number were recorded.

Data were analyzed with the Mann-Whitney  $U$  test (seizure number) and Student  $t$  test (seizure onset and durations). Data were considered significant if  $p < 0.05$ .

At the completion of the experiment animals were perfused with a buffered saline solution, followed by 10% buffered formalin. Brains were removed and processed for histologic study. Forty micrometer coronal sections of the brains were mounted on slides and stained with cresyl violet for verification of electrode and cannula placements. Histologic analysis of the brains of the NMDA-injected rats revealed that all electrodes were located within the dorsal hippocampus, whereas all cannulae were located in the lateral cerebral ventricle.

## Results

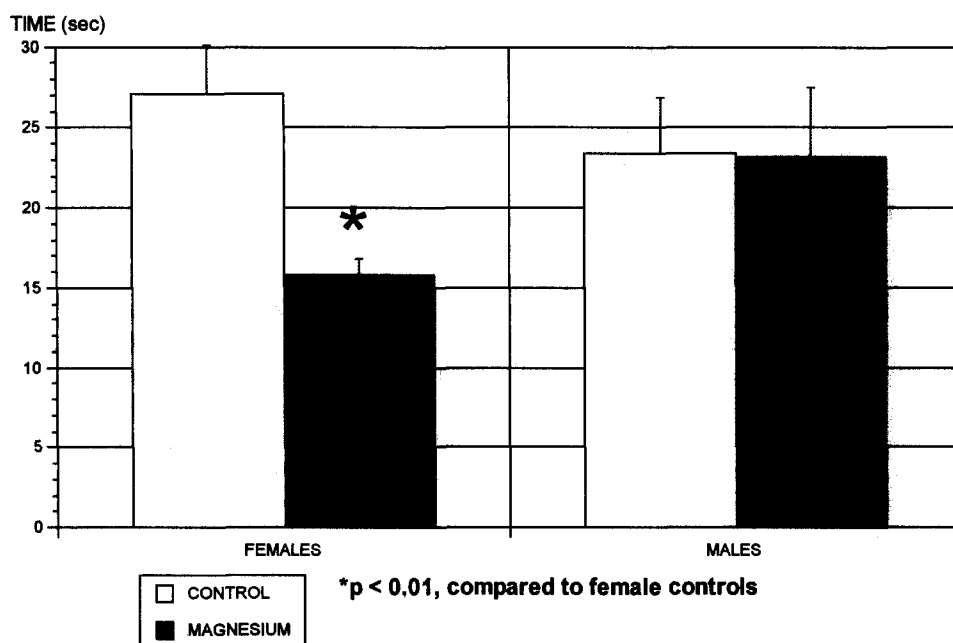
In saline solution-treated rats receiving the convulsant NMDA, females had a significantly reduced total

seizure duration ( $p < 0.01$ , Fig. 1) and total seizure number ( $p < 0.05$ , Fig. 2) compared with male rats. The duration of the first seizure was not affected by gender (Fig. 3). Time of onset to seizure activity was also not affected by sex (males  $24.4 \pm 8.0$  seconds, females  $31.7 \pm 7.2$  seconds).

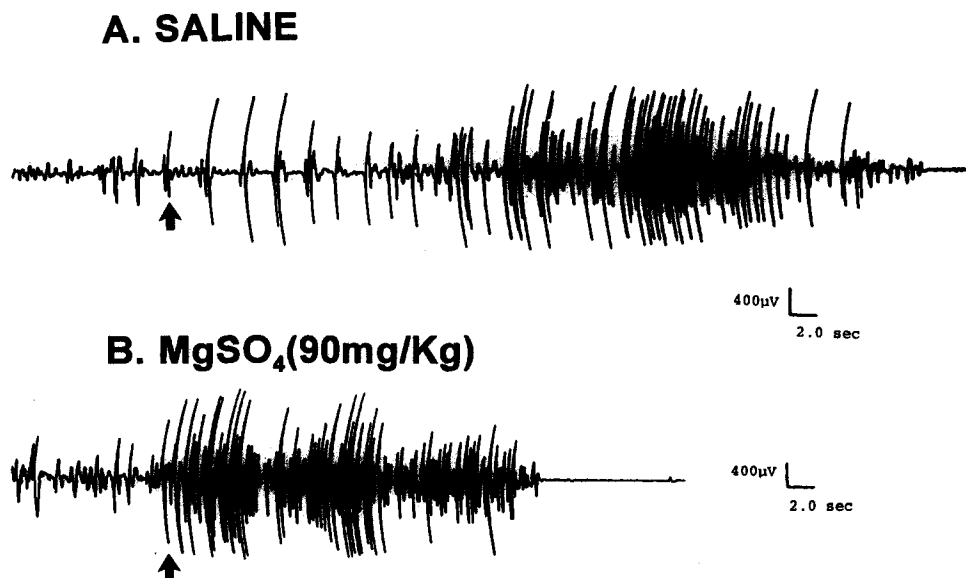
In male animals magnesium sulfate significantly reduced both total seizure duration ( $p < 0.05$ , Fig. 1) and total seizure number ( $p < 0.05$ , Fig. 2) compared with saline solution-treated controls. In females, on the other hand, magnesium sulfate significantly reduced first seizure duration ( $p < 0.01$ , Figs. 3 and 4) but did not affect total seizure duration or total seizure number over the 20-minute recording period.

The amplitude of the electroencephalographic recording was not affected by magnesium sulfate infusion in the males. However, NMDA injection depressed the electroencephalographic amplitude in both magnesium sulfate- and saline solution-treated rats (saline solution baseline  $99.6 \pm 5.9$   $\mu$ V, NMDA  $54 \pm 12.2$   $\mu$ V,  $p < 0.05$ ; magnesium sulfate baseline  $87.8 \pm 4.7$   $\mu$ V, NMDA  $40.2 \pm 12.1$   $\mu$ V,  $p < 0.01$ ).

As in the males, the amplitude of the electroencephalographic recording was not affected by magnesium sulfate infusion in the females but was decreased with central NMDA injection (saline solution baseline  $100.4 \pm 6.0$   $\mu$ V, NMDA  $47.8 \pm 6.3$   $\mu$ V,  $p < 0.01$ ; magnesium sulfate baseline  $98.5 \pm 6.1$   $\mu$ V, NMDA  $48.1 \pm 8.9$   $\mu$ V,  $p < 0.01$ ).



**Fig. 3.** Duration of first seizure in female and male rats given 5  $\mu$ g NMDA intraventricularly and either saline solution control or magnesium sulfate intravenously. There was no difference in duration of first seizure in female controls versus male controls. Magnesium sulfate treatment significantly reduced duration of first seizure in females but not in males (asterisk,  $p < 0.01$ , compared with female controls). Error bars, SEM.



**Fig. 4.** Representative EEG recordings of first seizure elicited in female rats given 5  $\mu$ g of NMDA intraventricularly. Magnesium sulfate ( $MgSO_4$ ) treatment reduced duration of first seizure. Arrow, Start of seizure activity.

#### Comment

Intracerebroventricular injection of NMDA readily induces characteristic behavioral seizure activity, including chewing, blinking, forelimb clonus, circling, jumping, and wild running. Electrical seizure activity in the

brain is represented on an electroencephalogram as periods of high-frequency, high-amplitude spiking referred to as an afterdischarge. This model of seizure activity has been used by other investigators to evaluate the effectiveness of various anticonvulsants in rats.<sup>9</sup> We

have previously reported a dose-response study of seizures induced by central NMDA injections, which demonstrated that 20  $\mu$ g of NMDA produces reliable seizure activity.<sup>10</sup>

In the current study we have shown that low-dose NMDA treatment results in significantly more severe seizure activity in males versus females as manifested by an increased number of seizures over 20 minutes and seizures of longer duration. Furthermore, we found that magnesium sulfate's anticonvulsant activity on NMDA-induced seizures was more pronounced in males than in females. Last, females seemed to have a resistance to low-dose NMDA-induced seizures not seen in the males.

No difference was observed in the time of onset to seizure activity between males and females in this study. Seizures occurred within seconds of intraventricular injection of NMDA, suggesting that the difference observed between male and female rats was not due to drug metabolism of NMDA. We have previously shown that the dose of magnesium sulfate used in the current study produces therapeutic blood levels in female rats and also produces an elevation in cerebrospinal fluid levels of magnesium.<sup>8</sup>

The differences in seizure activity between males and females is not likely to be due to the female estrus cycle. We, and others, have demonstrated that the estrus cycle does not affect seizure thresholds.<sup>11, 12</sup> This does not exclude the possibility that sex hormones can influence seizures. There are many studies that demonstrate gender effects on seizures,<sup>1-3</sup> suggesting that levels of sex hormones between males and females do influence seizure activity. In addition, there are reports of developmental differences in brain areas between males and females.<sup>13</sup>

It is also possible that the sex hormones progesterone and estrogen contribute to an anticonvulsant effect and modulate neuronal activity in rats. Progesterone and its metabolites act similar to the barbituates in that they modify  $\gamma$ -aminobutyric acid receptor binding characteristics and chloride channel activity.<sup>14</sup> Estrogens also have effects on the  $\gamma$ -aminobutyric acid inhibitory system. The enzyme glutamate decarboxylase, which is involved in  $\gamma$ -aminobutyric acid synthesis, is decreased in rats during treatment with estrogen.<sup>15</sup>

Studying seizure susceptibility in a clinical setting is difficult because of the presence of many confounding variables. However, an indication of the relationship between hormones and seizures is observed in the disease catamenial epilepsy.<sup>3</sup> A high ratio of estrogen to progesterone or lower levels of progesterone are thought to be causative in this disease. Moreover, enhanced electrical activity has been observed in electroencephalographic recordings done in women premenstrually versus midcycle.<sup>16</sup> How hormones relate to seizures in males is unknown.

Sex differences in NMDA receptor-mediated effects have been previously reported.<sup>17</sup> One study found that female rats responded to a peripheral injection of the noncompetitive NMDA antagonist MK-801 by becoming ataxic and hyperlocomotive, whereas male rats were not affected by the same dose.<sup>17</sup> Another study reported sex differences in the inhibitory effects of MK-801 on analgesic responses, showing increased male sensitivity compared with females.<sup>18</sup> Estradiol can regulate agonist binding to the NMDA receptor channel complex, as demonstrated with autoradiography.<sup>19</sup> Estrogen has also been shown to directly potentiate glutamate excitability.<sup>4</sup> On the other hand, estradiol application directly to female in vitro hippocampal brain tissue had no effect on neuronal excitability, whereas the hormone testosterone potentiated it.<sup>12</sup> Taken together, the data to date are controversial but suggest that seizure regulation by the NMDA receptor may be hormonally influenced.

In any event, it appears that females have a protective mechanism that prevents them from having severe seizure activity when given low-dose NMDA. A protective effect in females has been reported in studies of ischemia.<sup>20</sup> Females were found to have less ischemic vulnerability compared with males.<sup>20</sup> In addition, estrogen treatment given to males improved the neurological outcome after trauma-induced injury.<sup>21</sup> This protective effect of estrogen has been attributed to its lipid antioxidant properties.<sup>20</sup>

In the current study magnesium sulfate was more effective at reducing seizure activity over a 20-minute recording period in male rats than in females. However, the duration of the first seizure was effectively reduced in females but not in males. We have previously shown that magnesium sulfate reduces total seizure duration in female rats induced to convulse with high-dose NMDA (i.e., 20  $\mu$ g).<sup>11</sup> A central injection of 20  $\mu$ g of NMDA produces very intense behavioral and electrical seizure activity of long duration in female rats. Interestingly, seizures occur more frequently and are of longer duration in females than in males when tested with 20  $\mu$ g of NMDA (unpublished observation). It is possible that the anticonvulsant effects of magnesium sulfate are more readily observed with more severe seizures. However, magnesium sulfate did affect the first NMDA-induced seizure in females, although gender differences were not observed. This suggests that factors other than seizure intensity are involved.

The mechanism of magnesium sulfate's anticonvulsant effect may be related to a suppression of NMDA receptor activity. Magnesium is a physiologic antagonist of the NMDA receptor<sup>22</sup> and has been shown to inhibit activity of this receptor under a variety of experimental conditions.<sup>22, 23</sup> Additional actions of magnesium may include antagonism of calcium-regulated responses<sup>24</sup> and/or charge screening.<sup>25</sup>

In conclusion, the current study provides the first evidence that NMDA-induced seizure activity is gender dependent. Because most studies of seizure activity and anticonvulsant drug testing are done in males, including females in future studies, especially if the current findings are related to sex hormones and their modulatory effects, might be considered. Future investigation into the effects of hormones on excitatory amino acid-induced responses appears to be warranted.

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